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AI-POWERED MONITORING OF DAILY ROUTINES IN DEMENTIA USING WEARABLE DEVICES

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INTRODUCTION

Dementia affects more than 55 million people worldwide and is projected to double by 2050. Monitoring daily routine changes is essential for early detection of behavioral decline [1].

This project introduces an intelligent monitoring system capable of recognizing human activities such as mobility, sleep, personal hygiene, meals, and fall detection, while also identifying deviations from established routines. The system, Figure 1, integrates data from a wrist-worn and an in-ear wearable device [2], enabling continuous, non-invasive, and real-time behavioral monitoring in elderly individuals with dementia.

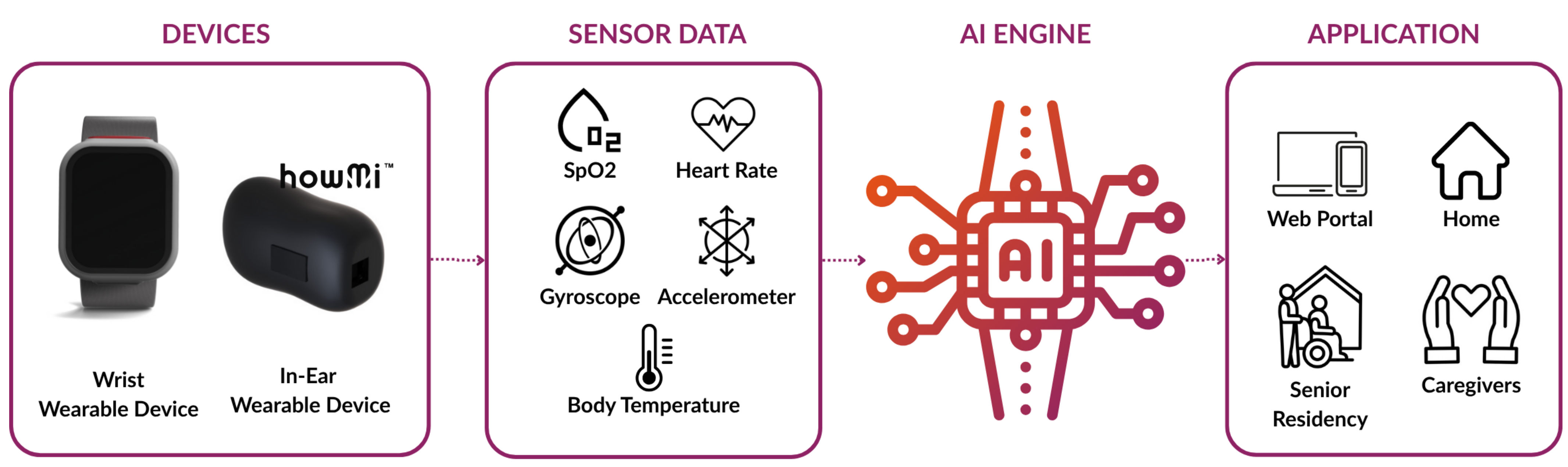

Figure 1. System architecture wearable wrist and ear devices, a local AI engine, and a web-based caregiver portal.

OBJECTIVES

- **Recognize daily activities** in elderly individuals using data from wearable sensors.
- **Learn typical routine patterns** to establish a personalized behavioral baseline.
- **Detect deviations from normal routines** that may indicate changes in health or behavior.
- **Trigger alerts** to caregivers when relevant anomalies or irregularities are detected.

DAILY ACTIVITY RECOGNITION

Publicly available wearable datasets were used to train machine learning and deep learning models to recognize daily activities such as **walking, sleeping, eating, and hygiene-related tasks** [3]. The performance of the following models was evaluated, with results shown in Figure 2.

Machine learning models:

Support Vector Machines (SVM)
Random Forest (RF)
K-Nearest Neighbors (KNN)
Logistic Regression (LR)
Gradient Boosting (XGBoost, LightGBM)

Deep learning models:

Convolutional Neural Networks (CNN)
Long Short-Term Memory (LSTM)
Gated Recurrent Units (GRU)

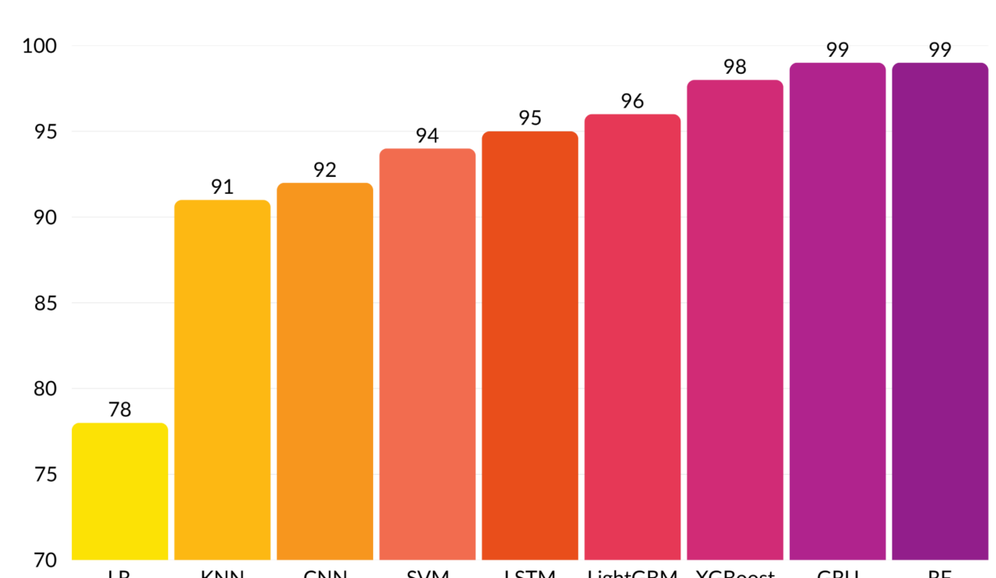

Figure 2. Model Accuracy Comparison for Activity Recognition.

CONCLUSIONS AND FUTURE WORK

This study demonstrated the potential of AI-based models to recognize daily activities and detect anomalies using only synthetic data. While results showed high recall for deviations, precision remains limited due to frequent false positives.

A major challenge is the lack of real-world data, particularly from elderly individuals or dementia patients, which hinders model validation and generalization. Future work will focus on:

- Collecting real-life datasets in realistic home settings;
- Reducing false positives through threshold calibration and ensemble methods;
- Personalizing alert**s** to support timely caregiver interventions.

These improvements are essential to ensure reliable, real-time monitoring in healthcare contexts.

ANOMALY DETECTION

To identify deviations from expected daily routines, this study applied **unsupervised deep learning approaches** that do not rely on labeled anomaly data. Two GRU-based models were trained exclusively on **synthetic normal behavior data** to learn routine patterns. Deviations were flagged based on **reconstruction or prediction errors**, which may reflect early signs of health deterioration such as agitation, inactivity, or disrupted sleep.

GRU Autoencoder:

- Learns to reconstruct normal behavior sequences.
- A day is labeled anomalous if >10% of its windows exceed the error threshold (95th percentile of training errors)

GRU Seq2Seq:

- Predicts the next activity in a 30-step sequence.
- Anomalies detected by comparing predicted vs. actual activities using a threshold: mean + 2 × std deviation of training error

As shown in Figure 3, anomaly detection involves a GRU-based sequence model followed by a threshold-based classification.

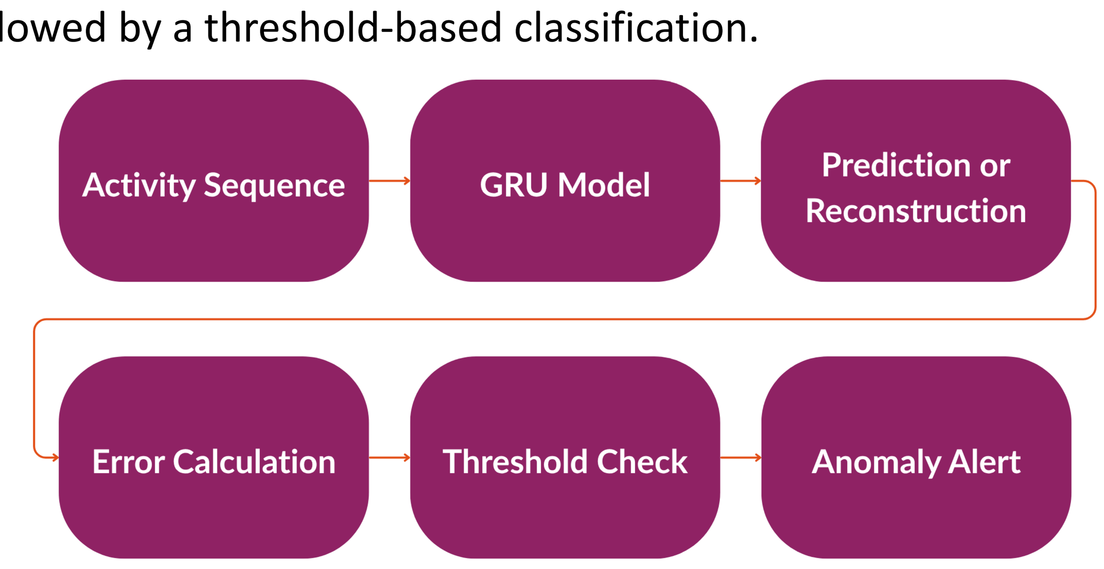

Figure 3. Anomaly Detection Workflow.

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